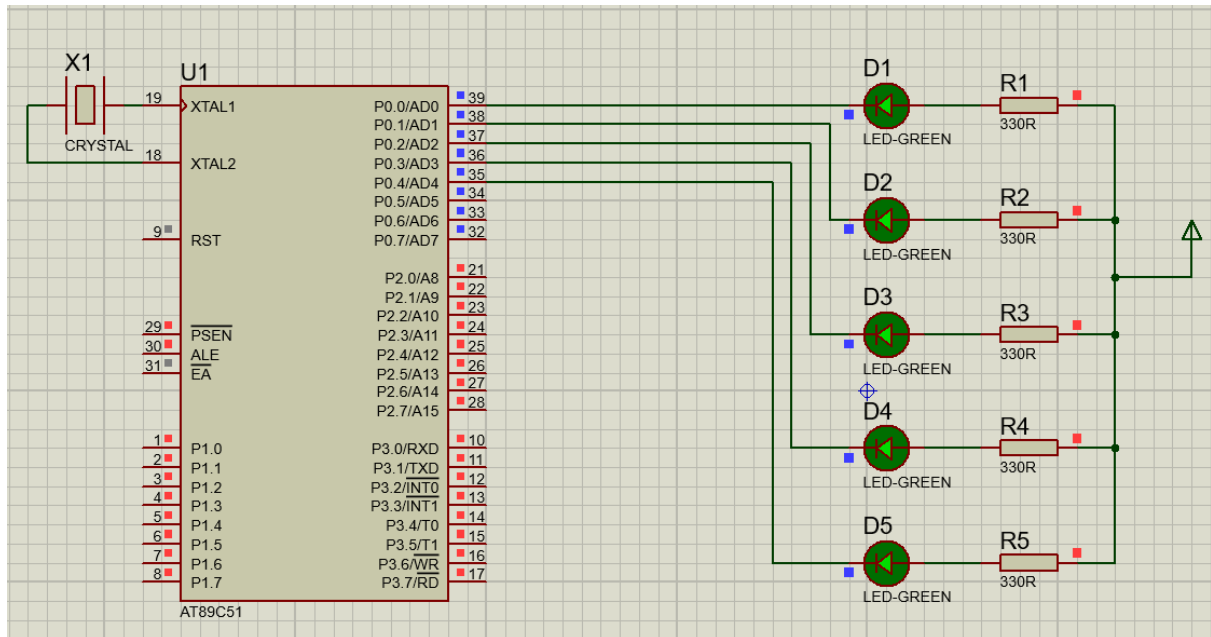


1 LED - Blinking

Aim: To write and execute a program to interface LEDs with the 8051 microcontroller & make them blink.



Apparatus Required:

- 8051 microcontroller
- LEDs
- Resistor (330Ω)
- Breadboard
- Connecting wires
- Keil μ Vision IDE

Theory:

- An LED is a semiconductor device that emits light when current flows through it.
- In controller interfacing:
 - LEDs are connected to the O/P pins of the microcontroller.
 - A resistor is used in series to limit current & prevent damage.
 - The LED glows when the O/P pin provides logic Low (0) or High (1).

Working Principle:

- If the LED is connected b/w Vcc & port pin → it glows when pin=0 (current sinking).
- If the LED is connected b/w port pin & GND → it glows when pin=1 (current sourcing).

Program Code:

```
#include <reg51.h>

// Standard hardware delay
void delay(unsigned int time)
{
    unsigned int i, j;
    for(i = 0; i < time; i++)
    {
        for(j = 0; j < 1275; j++); // Approx 1ms delay at 11.0592M
        Hz/12MHz
    }
}

void main()
{
    // Initialize ports to a known starting state
    P0 = 0x00;
    P2 = 0xFF;

    while(1)
    {
        // Toggle all bits on both ports simultaneously
        P0 = ~P0;
        P2 = ~P2;

        delay(50);
    }
}
```

Result: The LED was successfully interfaced with the 8051 microcontroller & made to blink as per the program.

Precautions:

- Always use a current-limiting resistor with an LED.
- Ensure proper power supply (5V).
- Avoid wrong polarity of the LED.
- Check all connections before powering ON.